Quiz 5: grading rubric

*6.47 After closing the switch in the circuit of Fig. P6.47 at $t = 0$, it was reopened at $t = 1$ ms. Determine $i_C(t)$ and plot its waveform for $t \geq 0$. Assume no energy was stored in either L or C prior to $t = 0$.

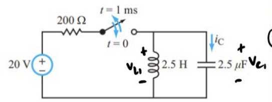

Figure P6.47 Circuit for Problem 6.47.

① $t = 0^-$

$i_{c_1}(0^-) = 0 = i_{L_1}(0^-)$ +1 $t < 0$

$V_{c_1}(0^-) = 0 = V_{L_1}(0^-)$

$V_{L_1} = L \cdot \frac{di_L}{dt}$

$V_{L_1} = L \cdot i_L'$

② $t = 0$

$i_L(0) = 0$

$i_{L_1}'(0) = \frac{V_{L_1}(0)}{L} = 0 = \frac{V_{c_1}(0)}{L}$

③ $0 < t < 1$ [ms] ⟵ ④ source tranformatio. $\frac{0\,[V]}{200\,[\Omega]} = 0.1$ [A]

time segment 1   +1 redraw

⑤ damping parameters:

$\alpha = \frac{1}{2RC} = \frac{1}{2 \times 200 \times 2.5 \times 10^{-6}} = 1000\ [s^{-1}]$

$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{2.5 \times 2.5 \times 10^{-6}}} = 400$ [rad/s]   +2 parameters

$\alpha > \omega_0 \Rightarrow$ overdamped

go to table 6-2

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⑥ get general solution: $i_{L_1}(t) = i_{L_1}(\infty) + A_1 e^{s_1 t} + A_2 e^{s_2 t}$

$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -83\ [s^{-1}]$ $i_{L_1}(\infty) = 0.1$ [A]

$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -1917\ [s^{-1}]$

$A_1 = \frac{i_{L_1}'(0) - s_2 [i_{L_1}(0) - i_{L_1}(\infty)]}{s_1 - s_2} = -0.1045$ [A]   +4 for s_1, s_2, A_1, A_2

$A_2 = \frac{i_{L_1}'(0) - s_1 [i_{L_1}(0) - i_{L_1}(\infty)]}{s_1 - s_2} = 0.0045$ [A]

⑦ $i_{L_1}(t) = [0.1 - 0.1045 e^{-83t} + 0.0045 e^{-1917t}]$ [A]

⑧ $i_{c_1}(t) = LC i_{L_1}''(t)$ ⟵ $\left[I = C \cdot \frac{dv}{dt}, \quad V_L = L \frac{di_L}{dt} \right]$ → $I = C \cdot \frac{d}{dt}\left(L \frac{di_L}{dt}\right)$ $V_C = V_L$

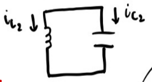
$C \cdot v_c'$

$i_{c_1}(t) = 2.5\ [H] \cdot 2.5\ [\mu F] \cdot \frac{d}{dt}\left[\frac{d}{dt}(0.1 - 0.1045 e^{-83t} + 0.0045 e^{-1917t})\right]$

$\boxed{i_{c_1}(t) = (-0.0045 e^{-83t} + 0.1045 e^{-1917t})\ [A]}$ +2 final $i_{c_1}(t)$ eq.

10 pts so far.

⑨ $t > 1\,[ms]$ time segment 2



+1 redraw $i_{L2} + i_{C_2} = 0$

$i_{L_2} = -i_{C_2}$ $\qquad v_{C_2} = v_{L_2}$ $\qquad i_{L_2} = i_{C_2}$

$$i_{C_2} = C \cdot \frac{dv_{C_2}}{dt} = C \cdot \frac{dv_{L_2}}{dt} = LC\frac{d^2 i_{L_2}}{dt^2} = -LC\frac{d^2 i_{C_2}}{dt^2}$$

$$i_{C_2} = -LC\frac{d^2 i_{C_2}}{dt^2} = -LC i_{C_2}''$$

$$i_{C_2}'' + \frac{1}{LC} i_{C_2} = 0$$

⑩ general form is similar to underdamped: $\left[i_{C_2}(t) = A_1 \cos \omega_0 t + B_1 \sin \omega_0 t \right]$

$$\omega_0 = \frac{1}{\sqrt{LC}} = 400\ [\text{rad/s}]$$ +1 for ω_0

⑪ get $i_L(t)$ + $v_c(t)$ for both time segments

$i_{L_1}(1\,[ms]) = 0.0045\ [A]$ (plug into ⑦)

$i_{L_2}(1\,[ms]) = -i_{C_2}(1\,[ms]) = -0.92A_1 - 0.39B_1$ ① (plug into ⑩)

two time segments

$v_{C_1}(t) = v_{L_1}(t) = L \cdot i_{L_1}'(t)$

$v_{C_1}(t) = L \cdot [0.1045 \cdot 83e^{-83t} - 0.0045 \cdot 1917e^{-1917t}]$

$v_{C_1}(1\,[ms]) = 16.86\ [V]$ +2 for correct V_{C1}

solve system:
① $-0.92A_1 - 0.39B_1 = 0.0045\ [A]$
② $389A_1 - 921B_1 = 16.86\ [V]$

+2 correct system of eq's.

$v_{C2}(t) = v_{L_2}(t) = L \cdot i_{L_2}' = -L\, i_{C_2}'$

$v_{C_2}(t) = -L \cdot \frac{d}{dt}(A_1 \cos 400t + B_1 \sin 400t) = 1000(A_1 \sin 400t - B_1 \cos 400t)$

⑫ $v_{C_2}(1\,[ms]) = 389A_1 - 921B_1$ ②

① ② solve system: $A_1 = 2.43 \times 10^{-3}\ [A]$ +1 correct valve A_1

$B_1 = -0.017\ [A]$ +1 correct valve B_1

$$\boxed{i_{C_2}(t) = (2.43 \times 10^{-3} \cdot \cos 400t - 0.017 \sin 400t)\ [A]}$$

+2 final eqvation